

→ Data :

Known facts that can be recorded and have an implicit meaning

Database :

Collection of related data

DBMS :

DB + Software

Mini-world :

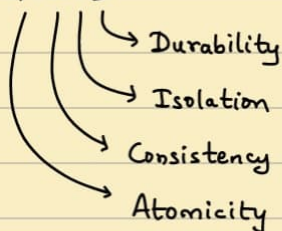
Some part of the real world about which data is stored in DB.

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Triggers and Stored Procedure → Not part of syllabus

Keywords : Entity , Attribute , Relations

- ACID



- Constraints , Operations , Datatypes

Database schema → Description | Catalogue

State → Snapshot

- Three - State Architecture

Two - tier Client - Server architecture

- DDL ← Data creation

DML ← Query Language

→ Create the following for an event of Infinium :

- (a) Tables
 - (b) Columns
 - (c) Datatypes
 - (d) Relationships
- } Database Schema

Database schema :

Infinium → Mini-world

e.g. Euphony :

STUDENTS form a band with other STUDENTS.

BAND has multiple roles - Singer, Pianist, Guitarist, Drummer, etc.

BAND gives auditions and proceed further if selected.

CLUB MEMBERS judge the auditions. They are also STUDENTS.

BANDS performs SONGs.

BANDS are given positions based on their performance

Band ID	Performer ID
int	int

Performer ID	Talent
int	varchar

Band ID	Selected/NOT
int	varchar

Band ID	Song ID
int	int

Band ID	Rank
int	int

Song ID	Song name
int	varchar

1 Band → n Performers

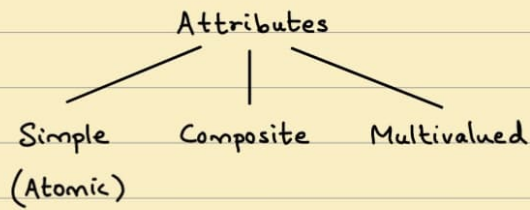
1 Band → n songs

1 Performer → 1 Talent

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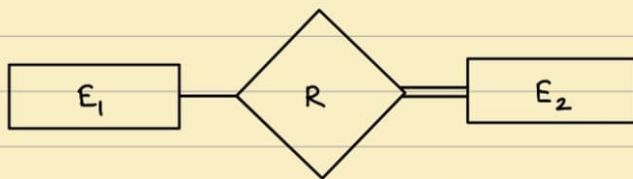
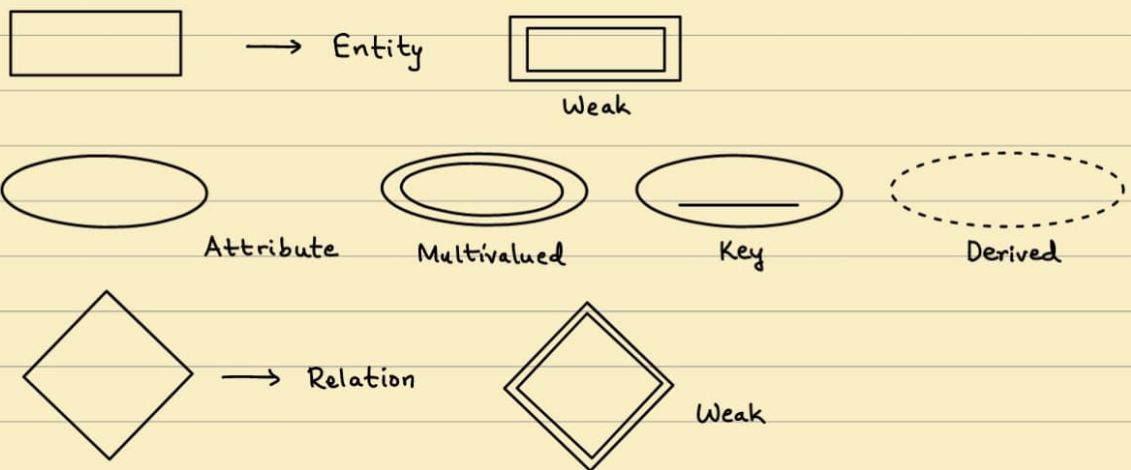
→ ER Model

- Database Designer : doesn't care how the data is captured.
- Entity : Basic concept of ER Model (Row)
- Attributes : Properties of entity (column)



Key Attribute → Each entity has a unique value

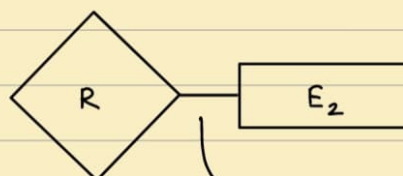
Representation : Entity. Attribute



→ Total participation of E_2 in R



→ Cardinality Ratio → 1:N for $E_1:E_2$ in R



→ (min-max) → Structural constraint

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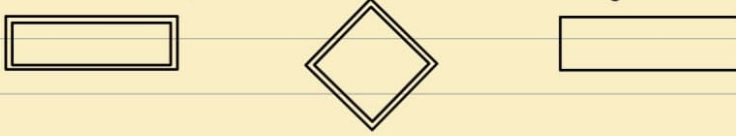
Relation types

Constraints

Recursive relations

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- Weak Entity $\rightarrow$ Depends on strong Entity



- Attributes of Relationship types

- Cardinality ratio

Participation constraint

- min-max notation

If you query often $\rightarrow$ Derived Attribute $\times$
Relationship type $\checkmark$

$\therefore$ Fetching is much faster

$\rightarrow$ Data Models:

(1) ER Models

(2) UML Class diagram

Entities $\leftrightarrow$ Classes

Relationships $\leftrightarrow$ Lines b/w classes

$\rightarrow$ DBMS:

Scripting Language

Embedded Approach

- Interface

- Utilities $\rightarrow$ Functions

- Cambridge Analytica - 3500 features stored about you

- Centralised DBMS Architecture

Client - Server Architecture $\rightarrow$ 3-Tier

→ Relational Model :

- Relation → Set of rows, i.e. tuple
Attribute → Column
→ Ordered set of values (1 tuple : 1 Row)
- Key : Uniquely identifies a row in a table
→ Can be set of attributes (or) attribute

- Schema : Table definition

$R(A_1, A_2, \dots, A_n)$
→ Attributes
→ name of the relation

- Domain :

- All possible column values
- Datatype constraint
Format constraint
- Developed from Attribute name

- Relation State :

$r(R)$ → Subset of Relation R

$$r(R) = \{t_1, t_2, \dots, t_n\}$$

$$t_i = \langle v_1, v_2, \dots, v_n \rangle$$

v_j → element of domain of A_j

$$r(R) \subset \text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n)$$

The tuples are not considered to be ordered.

The attributes are ordered. ← Self-Describing

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- Value : Atomic

value v_i of attribute A_i for tuple $t \leftarrow t[A_i] = v_i$

- Constraints :

- (1) Inherent (Implicit) : No duplicates
- (2) Schema-based (Explicit) : DB design
- (3) Application-based (Semantic) : By programmers

- Super Key : Attribute / set of attributes that uniquely identify all attributes in a relation
i.e. for any 2 distinct tuples t_1 & t_2 in $r(R)$: $t_1[SK] \neq t_2[SK]$
- Minimal Superkey $\rightarrow$ Key / Candidate Key
- Primary Key $\rightarrow$ Smallest of the candidate Keys
 $\rightarrow$ Cannot be NULL
- Foreign Key $\rightarrow$ Referential Integrity constraint

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$\rightarrow$ Keys

$\rightarrow$ Integrity Violation

- Domain constraint
- Key constraint
- Referential Integrity
- Entity Integrity

ON DELETE $\left\{ \begin{array}{l} \text{RESTRICT} \\ \text{SET NULL} \\ \text{CASCADE} \end{array} \right.$

$\rightarrow$ SQL :

AUTHORIZATION — Owner of Schema

Catalogue $\rightarrow$ Collection of Multiple Schemas